

Solving One-Step Equations with Rational Coefficients

Answer Key

Grades 6–7 Expressions and Equations

Quick Answers

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|-------|------------------|-----------|
| 1. 21 | 5. 12 | 9. 4 |
| 2. 15 | 6. -20 | 10. 8, 27 |
| 3. 12 | 7. 6 | |
| 4. 36 | 8. $\frac{4}{5}$ | |
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Common wrong answers

5. *If they got 33.3333: multiplied by the reciprocal five thirds instead of substituting three fifths*
 10. *If they got 18: substituted into the reciprocal three halves instead of two thirds*
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1. Tape diagram: one of three equal parts is 7.

Step 1. Read the diagram as an equation before computing. One of three equal parts of x is $\frac{1}{3}x$, and the diagram says that part is 7: $\frac{x}{3} = 7$.

Step 2. The undo of dividing by 3 is multiplying by 3, applied to both sides: $3 \cdot \frac{x}{3} = 3 \cdot 7$.
 $x = 21$

Check: $21 \div 3 = 7$, and the bar of 21 does split into three parts of 7.

2. $0.4x = 6$.

Step 1. The coefficient is a decimal, so the single operation done to x was multiplication by 0.4. Division undoes it.

Step 2. Divide both sides by 0.4: $x = \frac{6}{0.4}$. Multiplying top and bottom by 10 clears the decimal, giving $\frac{60}{4}$.
 $x = 15$

Sense check: 0.4 is less than 1, so taking 0.4 of the jar left fewer marbles than the jar held — the solution must be bigger than 6.

3. Tape diagram: two of three equal parts are 8, so $\frac{2}{3}x = 8$.

Step 1. Part (a): $x = 4$ would make one part $\frac{4}{3}$, and two parts would be under 3 — nowhere near 8. The 8 counts *two* parts, not the whole bar.

Step 2. Multiply both sides by the reciprocal $\frac{3}{2}$: $\frac{3}{2} \cdot \frac{2}{3}x = \frac{3}{2} \cdot 8$, so $x = \frac{24}{2}$.
 $x = 12$

Check on the diagram: each part is $12 \div 3 = 4$, and two parts make 8.

4. $\frac{x}{4} = 9$ ounces.

Step 1. One quarter of the bag is 9, so the coefficient is the unit fraction $\frac{1}{4}$ and its reciprocal is simply 4.

Step 2. Multiply both sides by 4: $x = 4 \cdot 9$.

$$x = 36$$

Check: a bar of 36 cut into 4 equal parts gives parts of 9.

5. **Testing Mia's claim that $x = 20$ solves $\frac{3}{5}x = 12$.**

Step 1. A check is a substitution, not a re-solve: put 20 in place of x and evaluate the left side only.

Step 2. $\frac{3}{5} \cdot 20 = \frac{60}{5}$, which is 12 — exactly the right side, so Mia is right.

$$12$$

Watch out: a student who reaches for the reciprocal here computes $\frac{5}{3} \cdot 20 \approx 33.33$ and wrongly rejects a correct answer. The reciprocal is for *solving*; a check uses the equation as written.

6. $-\frac{3}{5}x = 12$.

Step 1. Part (a): a negative coefficient times x gives a positive 12, so x must be negative. Deciding this first turns the sign into a prediction you can check.

Step 2. Multiply both sides by the reciprocal, sign included, $-\frac{5}{3}$: $x = -\frac{5}{3} \cdot 12 = -\frac{60}{3}$.

$$x = -20$$

Check: $-\frac{3}{5} \cdot (-20) = 12$, and the sign matches the prediction.

7. $1.25x = 7.5$ dollars.

Step 1. Part (a): the coefficient 1.25 is greater than 1, so each gigabyte costs more than a dollar and the number of gigabytes must be smaller than 7.5.

Step 2. Divide both sides by 1.25: $x = \frac{7.5}{1.25}$. Multiplying top and bottom by 100 gives $\frac{750}{125}$.

$$x = 6$$

Check: $1.25 \cdot 6 = 7.5$, and 6 is indeed smaller than 7.5, as predicted.

8. $\frac{5}{6}x = \frac{2}{3}$.

Step 1. Fractions on both sides change nothing about the method: the coefficient is $\frac{5}{6}$, so its reciprocal $\frac{6}{5}$ is what clears it.

Step 2. $\frac{6}{5} \cdot \frac{5}{6}x = \frac{6}{5} \cdot \frac{2}{3} = \frac{12}{15}$, and $\frac{12}{15}$ reduces by 3.

$$x = \frac{4}{5}$$

Check: $\frac{5}{6} \cdot \frac{4}{5} = \frac{4}{6} = \frac{2}{3}$. Note the solution is smaller than the right side, because the coefficient is less than 1.

9. $\frac{3}{8}x = 1.5$.

Step 1. Mixed forms are easiest in one form: part (a) rewrites 1.5 as $\frac{3}{2}$, so both sides are now fractions.

Step 2. Multiply both sides by the reciprocal $\frac{8}{3}$: $x = \frac{8}{3} \cdot \frac{3}{2} = \frac{24}{6}$.

$$x = 4$$

Check: $\frac{3}{8} \cdot 4 = \frac{12}{8} = 1.5$.

10. Challenge: Devon's error on $\frac{2}{3}x = 18$.

Step 1. Part (a) is a substitution, so evaluate the left side at Devon's value: $\frac{2}{3} \cdot 12 = \frac{24}{3}$. 8

Step 2. Part (b): 8 is not 18, so $x = 12$ is not a solution. Devon multiplied by the coefficient $\frac{2}{3}$ when the undo of multiplying by $\frac{2}{3}$ is multiplying by its reciprocal $\frac{3}{2}$.

Step 3. Part (c): $\frac{3}{2} \cdot \frac{2}{3}x = \frac{3}{2} \cdot 18 = \frac{54}{2}$. $x = 27$

Check: $\frac{2}{3} \cdot 27 = 18$. Note also that $\frac{2}{3}$ is less than 1, so the solution has to be *larger* than 18 — Devon's 12 was on the wrong side of 18 before any arithmetic was checked.